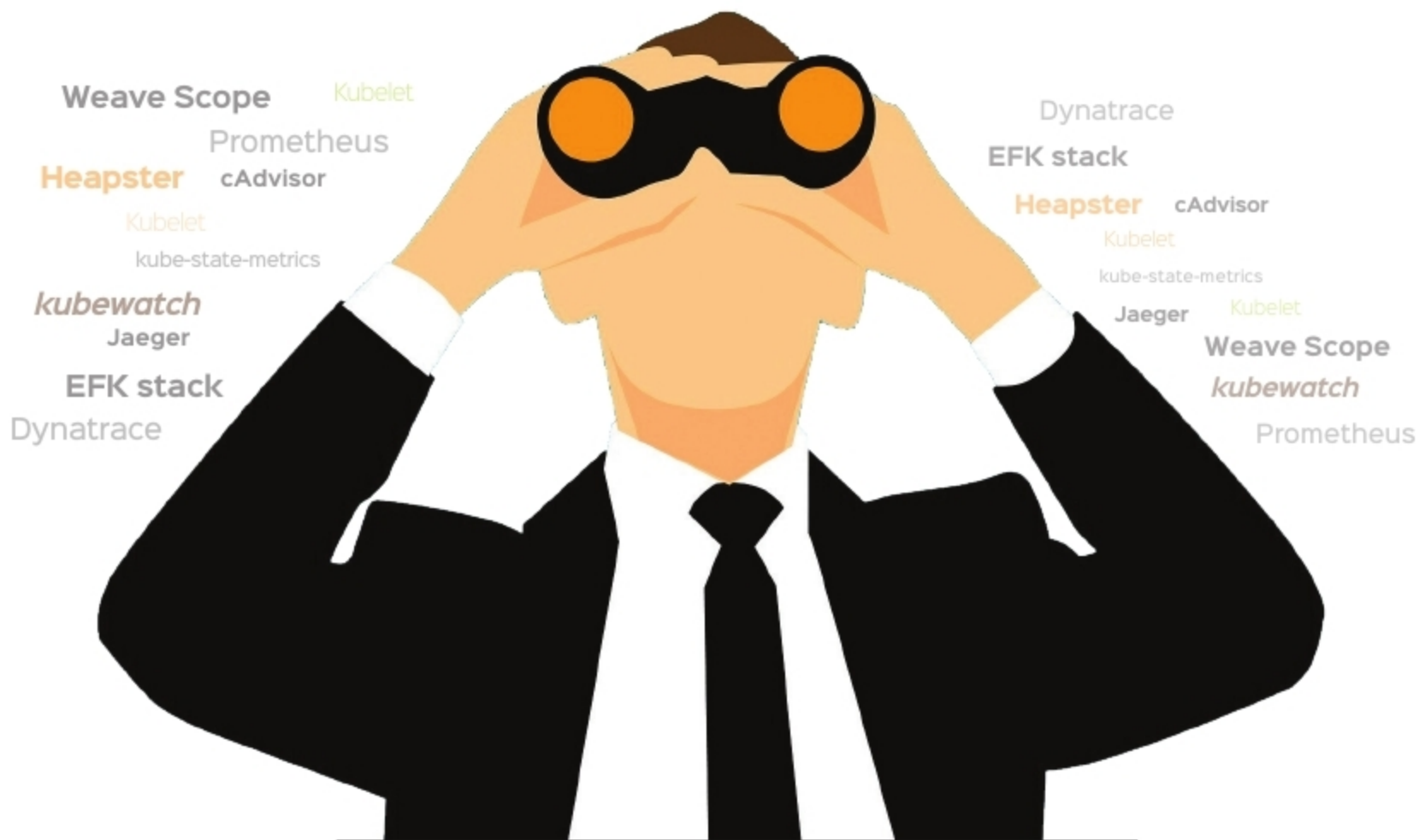


The Top Open Source Tools for Monitoring Kubernetes



While distributed computing and orchestration solve various problems when running applications, there is a price to be paid. A Kubernetes cluster appears as a single computer but, in reality, it is a connected set of multiple services and standalone modes. It is critical to update your monitoring strategies and open source tools to deal with the challenges of Kubernetes containers.



Container orchestration is a hot topic today in the field of software delivery. All organisations, from startups to big enterprises, depend upon containerised applications. However, monitoring of container orchestration is critical for the growth of IT, in any business. Open source tools are the answer to this. They help tackle the challenges associated with monitoring various environments through logs, metrics, interfaces and data collection.

The name Kubernetes is Greek for pilot or helmsman. True to its name, it

is a portable, extensible, open source platform for managing containerised workloads and services. It can be declaratively automated and configured. Kubernetes was built on the foundation of Google's experience of a decade-and-a-half with scalable production workloads.

The containerised approach offers the flexibility of running varied applications on a single platform without having to allocate different virtual hardware and operating systems, which streamlines the creation of application container images.

Kubernetes can run on a cloud based OS, offering environment portability and continuity of runtime with different applications as well as effective resource management for storage, etc. The allocation of resources like the CPU, storage and memory load, along with the creation of application container images, happens at build/release time. This decouples development and the operation of applications from the infrastructure. Kubernetes fits these containers on to the nodes for best resource utilisation (automatic bin packing).